

★New Years Mock AIME ★

NYMA Team

January 3 2024



- ❖ [ARML conventions](#) apply. These may be referred to throughout the test.
- ❖ This is a 15-question, 3-hour examination. All answers are integers ranging from 000 to 999, inclusive. Your score will be the number of correct answers. There is neither partial credit nor penalties for wrong answers
- ❖ Only scratch paper, graph paper, rulers, compasses, and erasers are allowed as aids. No calculators, smartwatches, phones, protractors, or computing devices are allowed. No problems on the exam will require the use of a calculator.
- ❖ Submit your answers [here](#)
- ❖ Enjoy! (mandatory)

Problem 1. On a circular track, two cross country runners start on diametrically opposite points and are both running counterclockwise. Both runners run at constant speeds and one runs twice as fast as the other. Whenever they meet on the track, the faster runner instantly reverses direction but the slower runner continues running in the same direction. If it takes them m minutes to meet for the first time and n minutes to meet for the 100th time, where m and n are relatively prime positive integers, find $m + n$.

Problem 2. Suppose $b > 1$ is a real number where $\log_5 (\log_5 b + \log_b 125) = 2$. Find $\log_5 (b^{\log_5 b}) + \log_b (125^{\log_b 125})$.

Problem 3. Alice has 9 distinct pairs of socks in her drawer. She picks 5 socks at random without replacement. Given that the expected value of the number of pairs of socks she draws is $\frac{m}{n}$ where m and n are relatively prime positive integers, find $m + n$.

Problem 4. Call a positive integer x with non-zero digits *fruity* if it satisfies $E(x) = 24$ where $E(x)$ is the number of trailing zeros in the product of the digits of x defined over the positive integers. Determine the remainder when the 30th smallest *fruity* number is divided by 1000. (Trailing zeros are consecutive zeroes at the end of a number)

Problem 5. Let ω_1 be a circle of radius 1 that is internally tangent to a circle ω_2 of radius 2 at point A . Suppose $\overline{AB}$ is a chord of ω_2 with length $2\sqrt{3}$ that intersects ω_1 at point $C \neq A$. If the tangent line of ω_1 at C intersects ω_2 at points D and E , find $CD^4 + CE^4$.

Problem 6. Suppose f is a function defined over the real numbers that only outputs real numbers, where for all real numbers x and y the following conditions hold:

$$\begin{aligned} f(1 - x^2) &= 1 - x^2 \\ f(x^2 + y^2 + 2) - f(x^2 + 1)f(y^2 + 1) &= f(x^2 + 1) + f(y^2 + 1) \\ f(y^2) &\geq 0 \end{aligned}$$

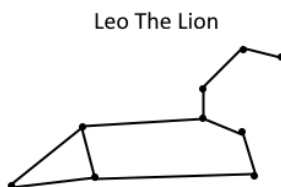
Then $\sum_{n=0}^8 f\left(n - \frac{1}{2}\right)$ can be written as $a\sqrt{b} - c$, where a , b , and c are positive integers and b is not divisible by the square of any prime. Find $a + b + c$.

Problem 7. Suppose S_n is the set of positive divisors of n , and denote $|X|$ as the number of elements in a set X . Let ξ be the set of positive integers n where $|S_n| = 2m$ is even, and S_n can be partitioned evenly into pairs $\{a_i, b_i\}$ for integers $1 \leq i \leq m$ such that the following conditions hold:

- ❖ a_i and b_i are relatively prime for all integers $1 \leq i \leq m$
- ❖ There exists a j where 6 divides $a_j^2 + b_j^2 + 1$
- ❖ $|S_n| \geq 20$

Determine the number of positive divisors $d \mid 24!$ such that $d \in \xi$.

Problem 8. Stars can fall in one of seven stellar classifications. The constellation Leo contains 9 stars and 10 line segments, as shown in the diagram, with stars connected by line segments having distinct stellar classifications. Let n be the number of valid stellar classifications of the 9 stars. Compute the number of positive integer divisors of n .



Problem 9. Let ABC be an isosceles triangle with $AB = AC = 4$ and $BC = 5$. Two circles centered at B and C each have radius 2, and the line through the midpoint of $\overline{BC}$ perpendicular to $\overline{AC}$ intersects the two circles in four different points. If the greatest possible distance between any two of those four points can be expressed as $\frac{\sqrt{a+b\sqrt{c}}}{d}$ for positive integers a, b, c , and d with $\gcd(b, d) = 1$ and a and c each not divisible by the square of any prime, find $a + b + c + d$.

Problem 10. Two complex numbers z_1 and z_2 are chosen uniformly and at random from the disk $0 \leq |z| \leq 1$. Suppose $z_1 = a + bi$ and $z_2 = c + di$, where a, b, c, d are all real numbers. The probability that

$$\frac{a^2 + b^2}{2abcd + a^2c^2 + b^2d^2} \leq \frac{a^2 + b^2}{-2abcd + a^2d^2 + b^2c^2} \leq 4$$

can be written as $\frac{p+q\pi-r\sqrt{s}}{t\pi}$, where p, q, r, s, t are positive integers, $\gcd(p, q, r, t) = 1$, and s is not divisible by the square of any prime. Compute $pqrst$.

Problem 11. Quadrilateral $ABCD$ has side lengths $AB = 24$, $BC = 40$, $CD = 56$, and $AD = 56$ with $\angle ADC = 60^\circ$. Point E is chosen in the plane such that $\overline{DE}$ is perpendicular to $\overline{DB}$, and points A , C , and E are collinear. Extend $\overline{BC}$ to intersect $\overline{DE}$ at point F , and $\overline{DC}$ is extended to point G such that line $\overline{BG}$ is perpendicular to $\overline{FG}$. Given that the area of quadrilateral $CGEF$ can be expressed as $a\sqrt{b}$, and a and b are positive integers and b is not divisible by the square of any prime, find the remainder when $a + b$ is divided by 1000.

Problem 12. Let $\Omega(n)$ be the function defined by $\Omega(1) = 0$ and $\Omega(np) = 1 + \Omega(n)$ whenever p is a prime number and n is a positive integer. Find the number of pairs of positive integers (x, y) that satisfy $x + y \leq 555$ and

$$\Omega(16xy^{x^2-1} + 4x + 4y^{x^2-1}x^5 + x^5) < 5.$$

Problem 13. A magician randomly lays out 13 distinct-valued cards a row, and then he passes his hand from left to right over the cards until his hand hovers over the least or greatest-valued card in the line. He removes that card. The number of cards he passes over is all the cards to the left of the removed card. If he keeps repeating this process on the row of cards until none remain. If x is the expected number of cards he will pass over, compute $x - \sum_{n=2}^{13} \frac{1}{n}$

Problem 14. If $f(n)$ denotes the number of divisors of 2024^{2024} that are either less than n or share at least one prime factor with n , find the remainder when

$$\sum_{n=1}^{2024^{2024}} f(n)$$

is divided by 1000.

Problem 15. Triangle ABC has side lengths $AB = 10$, $AC = 17$, $BC = 21$, and circumcircle ω . Let Ω with center O be a circle internally tangent to ω at A , and tangent to $\overline{BC}$ at D intersecting $\overline{AB}$ and $\overline{AC}$ again at points E and F , respectively. Suppose $\overline{AD}$ intersects ω again at M , and the tangent line to ω at A intersects line CB at X . If $\overline{XM}$ intersects ω at $N \neq M$, then the sum of the three circumradii of $\triangle XNO$, $\triangle BED$, and $\triangle DCF$ can be written as $\frac{m\sqrt{n}}{p}$, where m , n , and p are positive integers such that m and p are relatively prime, and n is not divisible by the square of any prime. What is $m + n + p$?

☆ Problem credits ☆

- Problem 1: arfekete
- Problem 2: arfekete
- Problem 3: Arcticturn
- Problem 4: eg4334
- Problem 5: arfekete
- Problem 6: arfekete
- Problem 7: blueprimes
- Problem 8: helloworld0
- Problem 9: Arcticturn
- Problem 10: SigmaPiE
- Problem 11: Arcticturn
- Problem 12: arfekete
- Problem 13: helloworld0
- Problem 14: arfekete
- Problem 15: r00tsOfUnity + EpicBird08